MILITARY SPECIFICATION

TRAINER, TEST SET TAU-62/E

1. SCOPE. This specification covers the requirements for a trainer designated as the TAU-62/E, hereafter referred to as the trainer, for use in special weapons training. This item is for classroom use and can be installed in a flight simulator.

2. APPLICABLE DOCUMENTS. The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-9107	Test Reports, Preparation Of
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery Of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-16	Electrical and Electronic Reference Designations
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Testing

DRAWINGS

Air Force

63A13108	Trainer, Test Set TAU-62/E
63D13086	Cable Assembly, Special Purpose, Electrical CX-9392/U

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(Copies of the above documents which are required by contractors in connection with specific procurement functions should be obtained from the Administrative Contracting Officer or as directed by the procuring activity.)

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.2 DESIGN AND CONSTRUCTION. The trainer shall be designed and constructed in conformance to drawing number 63A13108 and subsidiary drawings.

3.2.1 CABLE. The cable assembly shall conform to drawing number 63D13086. NOTE: This cable is used with, but not a part of, the trainer, and may be procured separately.

3.2.2 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria of MIL-D-70327 shall apply and new drawings shall be prepared as required.

3.2.3 FINISH. Finish of the trainer shall be in conformance to the engineering drawings.

3.2.4 DIELECTRIC STRENGTH. The electrical conducting parts shall be insulated from the case. The insulation shall be capable of withstanding repeated applications of at least one minute duration of voltages up to 50 volts rms at commercial frequencies.

3.3 PERFORMANCE. Satisfactory performance of the trainer shall consist of a visual indication that the proper set of numbers have been used during the operational sequence, and that the proper selection has been chosen on the AMAC (Arming and Control) knob.

3.4 ENVIRONMENTAL REQUIREMENTS. The trainer shall be capable of satisfactory performance after subjection to the following environmental conditions:

a. High Temperatures to 125°F.

b. Low Temperatures to -20°F.

c. Humidity incident to storage or service use in a damp climate.

d. Shock incident to transportation, handling and service conditions.

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3.5 IDENTIFICATION OF PRODUCT. The trainer, parts and assemblies shall be marked for identification in accordance with MIL-STD-130.

3.6 REFERENCE DESIGNATIONS. Reference designations used for marking within the unit shall be in accordance with MIL-STD-16. Markings shall not adversely affect the leakage path between conductors, or any other factor of performance. Reference symbols shall be employed to identify each part for its particular circuit application. The same reference symbol shall be used for identifying a particular part on all types of data, including drawings, diagrams, and spare parts lists prepared for the equipment. When practicable, markings shall be placed on the supporting structure adjacent to each replaceable part in such a position as to be readily reviewed.

3.7 WORKMANSHIP. The trainer shall be fabricated and finished in a manner such that criteria of appearance, fit and adherence to specified tolerances shall be observed. These criteria shall apply to all non-functioning parts as well as the functioning parts. Particular attention shall be given to marking of parts and assemblies, plating, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. Inspection and testing of the trainer shall be classified as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test units shall be representative of the production equipment. They shall be tested by a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

4.2.1.1 PREPRODUCTION REPORT. In the event that preproduction testing is performed at the contractor's facility, the contractor shall, at the completion of preproduction tests, prepare a preproduction test report. The preproduction test report shall be prepared in accordance with MIL-T-9107 and three complete copies shall be furnished the Government in accordance with instructions from the procuring activity.

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4.2 2 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described in "4.5 TEST METHODS" of this specification.

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of the following individual tests:

a. Examination of product.

b. Conformance to drawings.

c. Functional operational test.

4.3.1 INDIVIDUAL TESTS MAY CONTINUE. For operational reasons, individual tests may be continued pending the investigation of a test failure. Final acceptance of items on hand or produced later shall not be made until it is determined that the items meet all requirements of the specification.

4.3.2.2 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a test failure could indicate that defects exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 TEST CONDITIONS. Unless otherwise specified, test conditions shall be:

a. Temperature, $25^{o} \pm 10^{o}C$.

b. Relative Humidity, 80 percent or less.

c. Atmospheric Pressure, 28 to 32 inches of mercury.

4.5 TEST METHODS.

4.5.1 EXAMINATION OF PRODUCT. Each trainer shall be examined visually to determine compliance with all requirements of this specification for which tests are not specified. Failure to meet these requirements shall be cause for rejection.

4.5.2 CONFORMANCE TO DRAWINGS. Each trainer shall be inspected to determine conformance to the engineering drawings, subsidiary drawings and all specifications and standards referenced therein. Failure to conform to these criteria shall be cause for rejection.

4.5.3 FUNCTIONAL TEST. The functional tests shall be performed in the following steps in the sequence indicated:

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STEP	FUNCTION	RESULT
1	Connect CX-9392/U cable between the TAU-62/E student console and the instructor's console. Connect a 28v D.C. power source to the two spade lugs protruding from the connector plugging into the back of the student console. (NOTE: Positive lead must be connected to positive terminal on power supply)	With the cable connected, power shall be available to both the student panel and the instructor consoles. Once the 28v D.C. power source is applied, power shall be applied to the student panel lights.
2	Turn the AMAC knob on the student console from "Off" to "Safe."	Power shall be applied to TAU-62/E student console lights. "Safe" light on instructor panel shall light.
3	Move the control arm from the "Off-Safe" (O-S) position to the "Safe-Ground-Air" (SGA) position.	It shall now be possible to rotate the AMAC knob to the "Ground" and "Air" selections. NOTE: Prior to moving the control knob to the SGA position, it shall not be possible to turn the AMAC knob to the "Ground" or "Air" selection.
4	Operate the "Lamp Test" switch on the student console.	The "Status" and "Warning" lights on the student console shall light. The "Warning" light on the instructor panel will come on.
5	"Press-to-test" all lights on the instructor console.	All lights shall light with the exception of the "Status" light.
6	Rotate the AMAC knob from "Safe" to the "Ground" or to the "Air" selection.	The "Warning" lights on the student console and on the instructor console shall come on for 1.5 seconds plus or minus 0.5 seconds and then go out.

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		This will occur both in moving from "Safe" to "Ground" or "Air" selection and in returning to the "Safe" position. The "Safe," "Ground" or "Air" light on the instructor panel shall come on indicating the position at which the student has placed the AMAC knob.
7	Operate the "Malfunction" switch on the instructor console.	The "Warning" lights on both panels shall come on and stay on until the "Malfunction" switch is returned to the normal position.
8	Read the same code into the four encoder knobs on the Student console as appears on the encoder knobs of the Instructor console. Place the AMAC knob in any one of the three positions, "Safe," "Ground" or "Air," and rotate the enable knob from "Off" to "Enable."	The "Status" light on both the Student and the Instructor console shall light for 20 to 40 seconds and then go out. "Press-to-test" the "Status" light to determine that the coded device has enabled. NOTE: If an incorrect code has been set into the Student console, the "Status" light shall come on and stay on.
9	Turn the "Enable" knob to the "Off" position. Push the "Disenable" switch on the instructor console. After a period of 30 seconds, re-cycle through step No. 8.	Same as step number 8. NOTE: No re-cycling shall be possible unless the "Disenable" switch has been activated.
10	Disconnect the 28v D.C. power source from the TAU-62/E. Connect an ohmmeter between pin #3 of relay K1 and pin #3 of the Safe light. Set all Encoder knobs on the Student and Instructor consoles at zero. Read continuity. Rotate all eight encoder knobs through position 0 through 9 simultaneously. Read continuity at each position.	There shall be continuity. NOTE: Continuity will be broken when the first encoder knob is rotated to the next successive digit.

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Results other than those specified for each step shall be cause for rejection.

4.5.4 ENVIRONMENTAL TESTS. After subjection to the environmental tests, the trainer shall be subjected to the functional test. Failure to satisfy the requirements of the functional test shall be considered as failure to pass the environmental test and cause for rejection.

4.5.4.1 HIGH TEMPERATURE TEST. The trainer shall be subjected to the high temperature test of Method 501, MIL-STD-810, except that the high temperature shall be 125°F. After the trainer has returned to ambient conditions the functional test shall be performed.

4.5.4.2 LOW TEMPERATURE. The low temperature test shall be as described in Method 502, MIL-STD-810, except that the low temperature shall be -20°F. After the trainer has returned to ambient conditions, the functional test shall be performed.

4.5.4.3 HUMIDITY TEST. The humidity test shall be as described in Method 507, MIL-STD-810, except that the high temperature shall be 125°F. After the trainer has returned to ambient conditions, the functional test shall be performed.

4.5.4.4 SHOCK TEST. The shock test shall be as described in Method 516, Procedure VI, MIL-STD-810. At conclusion of the test, the trainer shall satisfy the requirements of the functional test.

5. PREPARATION FOR DELIVERY. Preparation for delivery shall be as specified in MIL-E-17555.

5.1 MARKING. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

TRAINER, TEST SET TAU-62/E

6. NOTES

6.1 INTENDED USE. Trainer, TAU-62/E is intended for use as a training device, either in classroom or flight simulator application.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be performed by the procuring activity.

c. Levels of preservation, packaging and packing.

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d. Provisions for separate procurement to Drawing No. 63D13086 of Cable Assembly, Special Purpose, Electrical CX-9392/U. This cable assembly is used with but not part of the trainer.

6.2.1 PREPRODUCTION TESTS. It is expected that the contract or purchase order will specify that a minimum of three units representative of the production trainer will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.3 PRODUCTION DELIVERY. No production trainer should be delivered prior to full approval of the preproduction tests.

6.4 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. ACCEPTANCE TESTS. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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